Physics - Graduate Level Assessment 13 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. By what factor would the heat transfer by radiation between the walls of a thermos bottle be increased if the reflectance of each wall were reduced from 0.9 to 0.3?

   (a) 11.3
   (b) 7.0
   (c) 13.7
   (d) 6.5
   (e) 12.5

2. Compared with the sound you hear from the siren of a stationary fire engine the sound you hear when it approaches you has an increased

   (a) volume
   (b) timbre
   (c) speed
   (d) duration
   (e) wavelength

3. How many calories are developed in 1.0 min in an electric heater which draws 5.0 amp when connected to a 110-volt line?

   (a) 5.0×10^3 cal
   (b) 15.3×10^3 cal
   (c) 10.5×10^3 cal
   (d) 12.7×10^3 cal
   (e) 7.9×10^3 cal

4. Calculate the mass of the electron by combining the results of Millikan's determination of the electron charge q and J.J. Thomson's measurement of q/m.

   (a) $6.7 \times 10^-17 kg$

   (a) $2.2 \times 10^-22 kg$

   (a) $3.0 \times 10^-32 kg$

   (a) $5.5 \times 10^-31 kg$

   (a) $1.76 \times 10^1 1$ kg

5. When an element ejects an alpha particle, the mass number of the resulting element

   (a) remains the same
   (b) increases by 4
   (c) reduces by 3
   (d) increases by 3
   (e) increases by 1

True / False (10 marks)

Answer True or False

6. True or False: The magnitude of the force on the particle is 5.0 N, as it relates to the rate of change of kinetic energy.

7. True or False: If a capacitor loses half its remaining charge every second and has charge q after five seconds, its initial charge was 16 q.

8. True or False: The magnification of the image of the filling in the tooth is 4 times the size of the object.

9. True or False: A force of 0.20 newton acting on a mass of 100 grams results in an acceleration of 5.0 m/sec^2.

10. True or False: Huygens' principle for light is primarily described by the behavior of particles, suggesting that light travels as discrete particles only.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the trajectory of a ballistic rocket launched from the equator at a 60°$anglewithaninitialspeed$

11. Discuss the dynamics of a head-on collision between a particle of mass m moving with velocity u_1 and a stationary particle of mass $2m$. How does the coefficient of restitution affect the final velocity v_1 of the first particle, particularly in maximizing kinetic energy loss? Include relevant equations and principles in your analysis.

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